

Gaussian noise. In addition, there is a text-processing component. These three components are described as follows:

1. **Autoencoder:** The input to the model is random noise, which is then reduced to a lower-dimensional latent space using a VAE architecture consisting of an encoder and decoder. During training, the encoder converts the sample into a lower latent representation, which is passed to the next block. The generated samples are denoised during inference and transformed back to their original dimensional latent space using reverse diffusion.
2. **U-Net:** The U-Net block uses ResNet to receive the noisy sample in a lower-dimensional latent space, compress it, and decode it back with less noise. The estimated noise residual from the U-Net output is used to construct the expected denoised sample representation.
3. **Text Encoder:** The text encoder is responsible for processing the prompt and transforming it into an embedding space. Stable Diffusion uses a frozen CLIP ViT-L/14 Text Encoder, similar to Google's Imagen.

Stable Diffusion Technical Details

Stable Diffusion v1 was initially trained on 256x256 images and fine-tuned on 512x512 images from a subset of the LAION-5B database. The model uses an autoencoder with a downsampling factor of 8, a UNet with 860M parameters, and a CLIP ViT-L/14 text encoder for diffusion. It is relatively lightweight and can run on a GPU with 10GB VRAM, or even less with float16 precision instead of float32.

The team has released several checkpoints, including sd-v1-1.ckpt trained for 237k steps on 256x256 resolution LAION-5B images and for 194k steps on 512x512 resolution images from LAION-5B with resolution $\geq 1024 \times 1024$. sd-v1-2.ckpt resumed from sd-v1-1.ckpt and was trained for 515k steps on laion-aesthetics v2.5+ subset with estimated aesthetics score > 5.0 , original size $\geq 512 \times 512$, and estimated watermark probability < 0.5 . Checkpoints sd-v1-3.ckpt and sd-v1-4.ckpt also resumed from sd-v1-2.ckpt and were trained for 195k and 225k steps respectively on the same subset as sd-v1-2.ckpt, but with a 10% dropping in the text-conditioning for improved classifier-free guidance sampling.

Stable Diffusion Model Performance

The Fréchet inception distance (FID) metric is commonly used to evaluate the quality of images generated by generative models. FID calculates the distance between the feature vectors of real and generated images. Imagen currently holds the best zero-shot FID score of 7.27 on the COCO benchmark,